

Gabriel M. Perry

Ph.D. Student in Computer Science | Feedback Systems Security | Mathematical Modeling |
Interim Secret Clearance
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Research Profile

Graduate researcher in computer science and control-oriented modeling, with work spanning robust feedback security, cyber-physical vulnerability analysis, system identification, algorithms, research software, teaching, and mentorship.

Research Interests

Robust control; cyber-physical systems security; feedback-system vulnerability analysis; system identification; Bayesian modeling; control-oriented machine learning; game theory; mathematical modeling; networked dynamical systems; algorithms and scientific computing.

Education

Brigham Young University <i>Ph.D. in Computer Science, expected April 2031</i>	Provo, UT 2026–2031
Brigham Young University <i>M.S. in Computer Science, defended April 2026</i>	Provo, UT 2024–2026

- Thesis: *A Masked-Perturbation Model for Destabilization Security Games in Feedback Systems.*
- Developed a theory of security games for feedback attacks in cyber-physical systems, emphasizing masked perturbations, destabilization, and attacker/defender structure.

Brigham Young University <i>B.S. in Computer Science, Data Science Emphasis; Minor in Mathematics, GPA 3.95, Cum Laude</i>	Provo, UT 2017–2024
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Research Experience

Research Assistant <i>Information and Decision Algorithms (IDeA) Labs, Brigham Young University</i>	2023–Present Provo, UT
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- Conduct research at the intersection of control theory, cyber-physical systems security, system identification, game theory, and computational modeling.
- Develop mathematical formulations and reproducible software for masked-perturbation security games, feedback-system vulnerability analysis, and critical-infrastructure applications.

RISC–ARLIS Summer Innovation Sprint, Team NY <i>GeoINT Methods for Supply Chain Anomaly Detection</i>	2025 Collaborative Research Sprint
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- Contributed to a collaborative geospatial-imagery machine-learning project for supply-chain anomaly detection, culminating in a technical repository and report.

Cyber-Physical Security Research Contributor <i>Attack Surface Modeling for Situational Awareness and Vulnerability Analysis of Power Systems</i>	2021–Present SBIR-linked Research
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- Contributed to research connected to DOE SBIR Phase I project DE-SC0021693 on power-system vulnerability analysis, in collaboration contexts involving Provo City Power, Oak Ridge National Laboratory, and KTH.

Selected Research Projects

- **Masked-Perturbation Model (M.S. Thesis).** Developed a feedback-security game framework for attacks and defenses that alter interconnection structure through masked perturbations; formalized destabilization-oriented attacker success and finite-action game reductions.

- **Bi-Directional Matrix Inverse Algorithm.** Designed and analyzed a matrix inverse method based on bi-directional graph search, resulting in a peer-reviewed IEEE CDC 2024 publication.
- **Bayesian System Identification (ongoing).** Developing Bayesian workflows to quantify model uncertainty and assess its downstream effects on control-relevant prediction and analysis.

Selected Research Software, Algorithms, and Technical Systems

- **masked-perturbation-model.** Research codebase for the M.S. thesis, with game formulations, transfer-function examples, numerical experiments, documentation, tests, and demonstrations.
- **modular_collatz.** Developed a reduced-memory algorithmic framework for Collatz dynamics, with an installable Python package, CLI, tests, and technical manuscript.
- **brent-scheme.** Built a research software package for tensor schemes in matrix multiplication; explored approximate recursive 3×3 schemes trading small numerical error for improved arithmetic scaling.
- **Agentic AI for Research Engineering.** Used agentic AI workflows to accelerate construction and professionalization of public research repositories, including package structure, tests, reports, and reproducible examples.
- **Web and React Development.** Built websites and React-based pages for communicating research, presenting technical work, and making public-facing materials more accessible.

Publications

- Gabriel M. Perry, Micah H. Olson, David Grimsman, and Sean Warnick. “A Matrix Inverse Algorithm based on Bi-Directional Graph Search.” *Proceedings of the 2024 IEEE 63rd Conference on Decision and Control (CDC)*, pp. 8926–8931, 2024.
- Tanner Sundwall, Gabriel M. Perry, David Grimsman, and Jonathan Sillito. “A bilevel approach to resource allocation for utility-based request-response systems.” *Proceedings of the 2025 American Control Conference (ACC)*, pp. 2268–2273, 2025.

Teaching, Mentorship, and Leadership

Graduate Teaching Assistant
Brigham Young University

2025–Present
Provo, UT

- Supported instruction in Game Theory and Theory of Predictive Modeling, with emphasis on mathematical reasoning, modeling principles, and conceptual explanation.

Undergraduate Teaching Assistant
Brigham Young University

2022–2023
Provo, UT

- Assisted students in Computation Theory, Discrete Structures, and Introduction to Computer Systems through proof guidance, programming support, and accessible technical explanation.

Chair, Operations Research and Engineered Systems Lab
Brigham Young University

2025–Present
Provo, UT

- Organize meetings and a training/lecture series on related research topics.
- Mentor undergraduate researchers in project development, research communication, and progress toward publishable work.

Public Technical Communication
Lectures, presentations, and research exposition

Ongoing
Online

- Maintain a public YouTube channel with thesis defense materials, research presentations, and lecture-style technical content.

Technical Skills

Programming	Python, MATLAB, C++, Java, JavaScript/React, L ^A T _E X; research-code packaging, testing, and reproducible computational workflows.
Machine Learning	PyTorch, TensorFlow, autograd-based modeling, Bayesian system identification, data-driven parameter learning, uncertainty-aware analysis.
Mathematics & Systems	Robust control, feedback systems, game theory, optimization, linear algebra, cryptography, stochastic modeling, networked dynamical systems.
Research Engineering	Scientific writing, notebook-to-package migration, technical reporting, agentic-AI-assisted software development, repository/documentation design.

Security Clearance

Interim Secret clearance.

References

Sean Warnick	Brigham Young University (Professor) sean@cs.byu.edu
David Grimsman	Brigham Young University (Professor) grimsman@cs.byu.edu
Brian Brown	Brigham Young University (Ph.D. Candidate) bcbrown365@gmail.com